Touching the Virtual: Multi-Touch Tangible Interfaces for Virtual Production

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ABSTRACT

This project expands the original virtual production tool, ***CineDesk,*** with a tangible controller on a touch table. It combines intuitive touch input with a peripheral (*Spacemouse*) to help 3D environment designers and animators plan staging, refine layouts, and create storyboards more efficiently. Early results show that combining both inputs reduces iteration time and improves team collaboration.





fig.1 Collaborative layout discussion using CineDesk's touch interface.

INTRODUCTION

CineDesk is a virtual production tool that helps teams plan 3D scenes more efficiently. It uses an intuitive, multi-modal interface with a large shared display to support real-time collaboration and fast camera setup.

- **Accessible:** Touch interface lets non-technical users adjust layouts easily
- **Collaborative:** Multiple users can interact at once, both physically and virtually
- **Fast Iteration:** Teams can visualize changes instantly and reduce setup time

RESEARCH OBJECTIVES

- **Design** a collaborative interface combining touch and tangible input
- **Compare** the hybrid method to traditional workflows in speed and efficiency
- **Evaluate** whether the SpaceMouse improves intuitiveness and usability

METHODS

- **Layered UI Architecture:** The UI system is structured into three primary layers:
  - 3D Scene Layer
  - Movable Widgets Layer
  - Widget Detail Layer

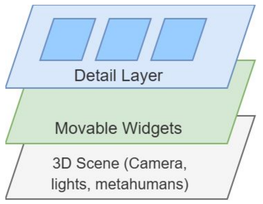


fig.2 CineDesk UI Architecture

- **Dual-Input Integration:** The methodology involves the simultaneous integration of direct touch interaction and a 6-Degrees-of-Freedom (6DoF) Spacemouse to provide both high-level staging and precise camera navigation.
- **Real-Time Rendering Pipeline:** Using the *Off World Live* plugin, the system streams CineCamera views via Spout output to provide a high-fidelity, real-time preview for the creative team.

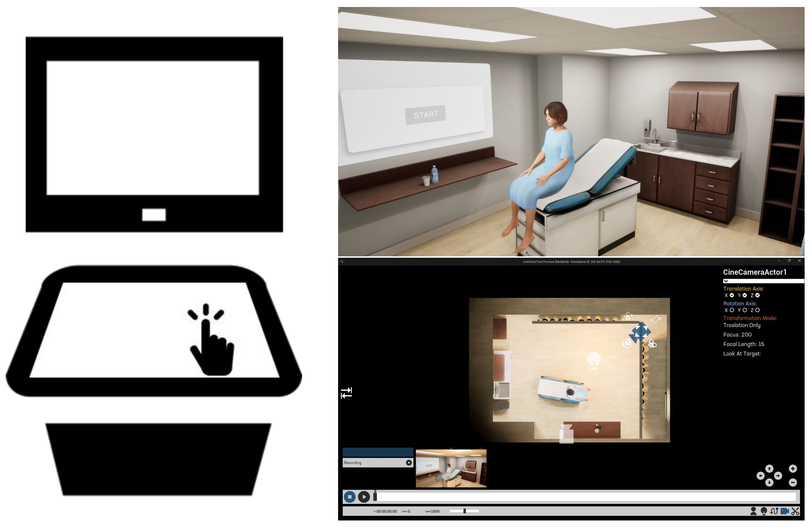


fig.3 Top-down layout view and 3D preview interface used for planning and staging.

RESULTS

- Users completed layout tasks faster with the dual-input system
- Participants reported improved camera control and ease of use
- Teams iterated more quickly and collaborated more fluidly

DISCUSSION

- **Tangible vs. Digital Control:** The results suggest that the Spacemouse provides a level of control and immediacy for camera setup that exceeds traditional digital-only inputs.
- **Collaborative Dynamics:** The system reduces technical friction and supports for more democratic participation in the pre-viz process.
- **Workflow Efficiency:** The combination of a top-down touch view for layout and a real-time 3D preview allows for two perspective simultaneously, which is critical for rapid storyboarding

CONCLUSION

Our findings suggest that hybrid touch and tangible input enhances both collaboration and control in virtual production. The dual-input system reduces iteration time and improves user experience for staging and camera setup.

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